

Overtune 3.2 (ot3) — System Installation Guide

NATIVE DEPLOYMENT, IN-APP AUTO-UPDATER, AND PORTABLE SHAREWARE BUNDLES

1. System Overview & Hardware Requirements

Overtune 3 (ot3) is a high-precision digital filter design and live audio simulation workstation built on pure C++20 and Qt 6. It features automated Stage 2 compliance mathematical audits, sub-millidecibel magnitude precision, and SIMD-accelerated cascaded biquad evaluation.

- **Linux Requirements:** Any modern 64-bit Linux distribution (Ubuntu 22.04+, Fedora 38+, Arch Linux, Debian 12+) with GLIBC 2.34+ and Qt 6.4+ libraries.
- **Windows Requirements:** Windows 10 (64-bit, 1809+) or Windows 11 (x64) with Microsoft Visual C++ 2022 Redistributable.
- **Display:** Minimum recommended display resolution of 1280 × 800 with HiDPI display scaling support.

2. Linux Installation & Execution

Overtune 3 provides three straightforward deployment methods on Linux platforms:

Method A: Native Automated Setup (install_linux.sh)

The interactive installer integrates **ot3** into your desktop environment, creates desktop and application menu entries, registers icons, and links the CLI binary:

```
# User-level installation (No root required - installs to ~/.local/share/
overtune3)
./scripts/install_linux.sh

# System-wide installation (Installs to /opt/overtune3 and /usr/local/bin)
sudo ./scripts/install_linux.sh
```

Once completed, launch Overtune 3 from your application menu or directly run **ot3** from any terminal.

Method B: Standalone Portable Shareware Bundle

For zero-install portable deployment (e.g., from a USB drive or local directory):

```
tar -xzf overtune3-linux-x86_64.tar.gz
cd overtune3-linux-x86_64
./ot3.sh
```

Method C: Building from Source

```
./build.sh                # Builds optimized release binary in ./build/bin  
    /ot3  
./build/bin/ot3           # Direct launch
```

3. Windows Installation & Execution

Overtune 3 provides comprehensive native Windows deployment options designed for both standard desktop users and power users:

Method A: Standalone Portable Shareware Package (Zero Install)

For instant, portable execution without administrative rights or registry modifications:

1. Extract `shareware\windows\overtune3-windows-x64.zip` (or copy the `overtune3-windows-x64` directory).
2. Double-click `ot3.bat` or `run.bat` to launch the application immediately.
3. Alternatively, navigate to `bin\` and execute `ot3.exe` directly.

Method B: Automated Native PowerShell Installer (`install_windows.ps1`)

To fully integrate Overtune 3 into Windows (Start Menu, Desktop, Apps list, and in-place auto-updater):

```
# Standard user or Administrator execution:
powershell -ExecutionPolicy Bypass -File .\shareware\windows\install_windows.ps1
```

Supported Options & Switches:

- `-InstallDir <path>`: Target folder (defaults to `%LOCALAPPDATA%\Programs\Overtune3`).
- `-NoDesktop`: Skip creating the Desktop shortcut.
- `-NoStartMenu`: Skip creating the Start Menu shortcut.

What the installer accomplishes:

- Copies the complete application bundle including all required Qt 6.7 runtime libraries, QML engines, multimedia backends, and assets.
- Creates the **Desktop** shortcut (`Overtune 3.lnk`) and **Start Menu** entry (`Programs\Overtune 3`).
- Registers **Overtune 3 Studio** in Windows Settings → *Installed Apps* with an uninstallation hook.
- Prepares `overtune_update.bat` for atomic background updating.

Method C: Building from Source on Windows (`build.bat`)

If building from repository sources using Visual Studio 2022 and Qt 6:

```
# One-click configure, compile, and runtime deployment:
.\build.bat

# Direct launch:
.\run.bat
```

This invokes CMake with the Visual Studio 17 2022 generator, compiles the pure C++20 DSP engine and Qt 6 GUI in parallel Release mode, and runs `windeployqt` to assemble all runtime DLLs and QML dependencies.

Method D: Modern NSIS Setup Executable

To generate or run a standard Windows Setup wizard:

```
# Compile NSIS installer (requires makensis):  
cd shareware\windows && makensis installer_windows.nsi
```

Run `Overtune3-Setup-x64.exe` interactively or silently with `/S`.

Windows Uninstallation

To remove Overtune 3:

- Open **Windows Settings** → **Apps** → **Installed Apps** → **Overtune 3 Studio** → **Uninstall**.
- Or run: `powershell -ExecutionPolicy Bypass -File "%LOCALAPPDATA%\Programs\Overtune3\uninstall.bat"`
- Or double-click: `"%LOCALAPPDATA%\Programs\Overtune3\uninstall.bat"`

4. In-App Auto-Updater & In-Place Atomic Updates

Overtune 3 features a unified OLED modal updater window matching the What's New and Stage 2 Verifier interfaces. It handles live patch updates seamlessly:

- **Accessing Updates:** Select **Help** → **Check for Updates...** or click the **Installer / Update Ready** badge in the titlebar.
- **Linux Atomic Patching:** Leverages POSIX atomic `unlink/rename` semantics. The running application binary is replaced smoothly in place without requiring process termination before write.
- **Windows Detached Updater:** Bypasses Windows executable file-locking by spawning a detached helper script (`overtune.update.bat`) that waits for the parent process PID to exit cleanly, replaces `ot3.exe` with `ot3.exe.new`, and restarts the updated studio application.
- **Cryptographic SHA-256 Audit:** All downloaded payload chunks are validated against release cryptographic checksums before application.

5. Command-Line Options & Essential Shortcuts

Option / Key	Type	Functionality
ot3 --install	CLI Flag	Launch directly into the native System Installation manager.
ot3 --update	CLI Flag	Query release server and display available version patches.
Ctrl + 1 .. 6	Keyboard	Navigate directly between Design, Analysis, Simulation, Export, Docs, Settings.
Ctrl + 0	Keyboard	Auto-scale and reset frequency, pole-zero, and impulse response plots.
F5 / Space	Keyboard	Start or pause the live real-time audio and DSP simulation engine.
Ctrl + T	Keyboard	Toggle between OLED Dark and Studio Light visual themes.

For support, bug reports, and full DSP mathematical derivations, refer to the in-app documentation page or open the `theory_and_math.pdf` manual.